

EXTRACTION — Enemy AI: Gap Closure & In-Engine Setup

Status: Action plan. Written 2026-06-10 on AI-Companion-Prototype.

Companion to: enemy_gameplay_as_built.md (§11 is the evidence behind every gap here), enemy_design.md, enemy_code_plan.md, enemy_test_levels.md (test-level layouts).

Scope: every gap between the design doc and what runs today, how to close each one, the complete in-engine setup checklist, and the full asset list. Ordered by what blocks the first playtest.

Who does what: [AGENT] = the NeoStack editor loop can do it autonomously (DA edits, BP graph wiring, actor placement). [HUMAN] = needs a person (mesh/animation/VFX choice and import, aesthetic judgement). [CLI] = C++ work for a Claude Code session. Many [HUMAN] rows become [AGENT] once the asset itself exists in the project.

1. Priority overview

P0 — blocks the first meaningful playtest. **P1 — blocks it *feeling* right. P2 — design-completeness, mostly C++.**

#	Item	Pri	Who	Effort
1	Bark lines: add GrenadeOut + Suppressing to DA_Barks_Grunt	P0	AGENT	2 min
2	DA_Enemy_Grunt DisplayName = "Grunt" (blank bark labels)	P0	AGENT	1 min
3	Sniper weapon: DA + BP child, assign to DA_Enemy_Sniper	P0	AGENT	15 min
4	Shield: real mesh + relative offset (cube currently blocks 360°)	P0	HUMAN mesh, AGENT wiring	30 min
5	Sniper laser beam bound to OnLaserChanged	P0	HUMAN FX, AGENT wiring	1–2 h
6	Grenade landing indicator bound to OnGrenadeTelegraph	P0	HUMAN FX, AGENT wiring	1–2 h
7	Meshes + ABP + physics asset + WeaponSocket on the 6 new BP children	P0	HUMAN	1–3 h
8	Spawn zones placed + DirectorConfig hookup + mission-phase triggers	P0	AGENT (placement intent from human)	30 min
9	SquadId on placed groups (incl. one officer-led squad)	P0	AGENT	10 min

#	Item	Pri	Who	Effort
10	NavMesh + stand-height cover slots over new arenas	P0	HUMAN/AGENT	30 min
11	Suppressed player weapon variant (stealth loop is untestable without it)	P0	AGENT	15 min
12	Ragdoll physics asset sanity check on the enemy mesh	P0	HUMAN	10 min
13	Hit-react / suppression-flinch / takedown montages bound to delegates	P1	HUMAN	2–4 h
14	Heavy plate-break visual on OnPlateBroken	P1	HUMAN FX	1 h
15	Rusher melee swing on OnMeleePerformed	P1	HUMAN	30 min
16	Grenade visuals: body mesh, trail, explosion FX + SFX on BP_EnemyGrenade	P1	HUMAN	1–2 h
17	Bark SFX per line (subtitle-first shipped; audio optional)	P1	HUMAN	open
18	Heavy LMG weapon variant (identity currently = burst length only)	P1	AGENT	15 min
19	Per-archetype bark sets (all seven share DA_Barks_Grunt)	P1	AGENT	30 min
20	Delete AEnemyBase.h/.cpp (permission-blocked in autonomous run)	P1	CLI	5 min
21	Grenade blasts: AI-hearing noise + suppression + morale event	P2	CLI	small
22	Officer aura: passive morale-floor raise + squad-only filter	P2	CLI	small
23	Morale events: "outnumbered" (loss), "reinforcements arrived" (gain)	P2	CLI	small
24	Lighting factor in sight fill	P2	CLI	medium
25	Side-aware "Flanking left/right!" bark	P2	CLI	small
26	Post-search "edgy" lingering state	P2	CLI	small
27	Rusher suppression response — confirm "fearless" intent or add a read	P2	decision first	—

2. Gap-by-gap closure plan

2.1 — P0 blockers

Gap 1+2: silent telegraphs + blank speaker labels (data-only, do these first)

Two bark types were never authored (code requests them; the subsystem silently drops types with no lines), and the grunt's DisplayName is empty.

1. Content Browser: /Game/Enemy/AI/DA_Barks_Grunt — open.
2. Barks map: add key Grenade Out — Lines: "Grenade out!", "Frag going in!" — CooldownSeconds 6.
3. Barks map: add key Suppressing — Lines: "Suppressing — move up!", "Covering fire!" — CooldownSeconds 6.
4. Save. Open /Game/Enemy/AI/DA_Enemy_Grunt — Identity, DisplayName = "Grunt". Save.

Done when: grenadier wind-up prints "Grenade out!" in the bark feed; overwatch suppressor prints "Suppressing — move up!"; grunt barks show "Grunt" as speaker.

Gap 3: sniper fires a 25-damage rifle

A 2-second-telegraphed quarter-health-bar poke doesn't control sightlines. Give him his own weapon.

1. Duplicate /Game/Core/Weapons/DA_AssaultRifle to DA_SniperRifle.
2. Set: BaseDamage 90 (head $\times 2.0$ = lethal on a 100 HP target; body = brutal-but-survivable — tune vs final player HP), MagazineSize 5, NoiseLoudness 1.0, NoiseRange 3500.
3. Duplicate /Game/Core/Weapons/BP_Rifle to BP_EnemySniperRifle; set its WeaponData = DA_SniperRifle.
4. Open /Game/Enemy/AI/DA_Enemy_Sniper — Weapon, WeaponClass = BP_EnemySniperRifle. Save.

Note: the sniper task fires exactly one round per telegraph (StartFiring+StopFiring) — fire rate doesn't matter, damage does.

Done when: a telegraphed body hit takes ~90% of a health bar; a head hit kills.

Gap 4: shield is an engine cube blocking 360°

The shield blocks by physical interception, and the runtime-created component sits at the mesh origin — until the mesh is replaced and pushed forward, the carrier is invulnerable from every angle and the archetype's counter (flanking) doesn't exist.

1. Import or pick a riot-shield static mesh (~120cm tall $\times$ 50cm wide; an engine/marketplace placeholder is fine if the pivot is on the grip side).
2. /Game/Enemy/AI/DA_Enemy_Shield — Shield, ShieldMesh = the new mesh. Save.
3. Open /Game/Enemy/BP_Enemy_Shield — Event Graph: add event OnBoltOnComponentsReady (BlueprintImplementableEvent on AEnemyCharacter — it fires right

after the component is created at possess).

4. From it: `GetShieldComponent -> SetRelativeLocation (X=45, Y=0, Z=40) -> SetRelativeRotation` (yaw so the face points along actor forward). Adjust in PIE until side/rear shots land on the body.

5. Compile, save.

Done when: frontal shots visibly hit the shield (no damage), a shot from 90° hits the carrier, a grenade pops the shield (400 shield HP, radial ×2) and he drops to cover-fighting.

Gap 5: sniper laser is invisible

`OnLaserChanged(bool bOn, AActor* Target)` broadcasts at telegraph start/cancel/shot — nothing is bound. The design's one hard rule for this archetype ("never a sniper without it") is currently violated.

1. Create a Niagara beam system `NS_SniperLaser` (red beam, two user params: Start, End — the engine beam template works).
2. `/Game/Enemy/BP_Enemy_Sniper` — Event Graph: `GetSniperTelegraphComponent -> Bind Event to OnLaserChanged`.
3. On `bOn=true`: spawn `NS_SniperLaser` attached to the weapon muzzle; on Tick-while-active (or a 0.05s timer), set End = Target eye location. On `bOn=false`: deactivate/destroy.
4. Optional but recommended: a quiet rising aim SFX during the 2.0s telegraph.

Done when: every sniper shot is preceded by 2 full seconds of visible beam, and breaking LOS or suppressing him kills the beam without a shot.

Gap 6: grenade flush has no visual telegraph

`OnGrenadeTelegraph(FVector Landing, float TimeToImpact)` and `OnGrenadeCancelled` are exposed on the grenadier component — unbound. With Gap 1 fixed you get the bark; this adds the ground marker.

1. Create a decal material (or Niagara ring) `M_GrenadeWarning` — red ring, ~350 radius (the blast radius).
2. `/Game/Enemy/BP_Enemy_Grenadier` — Event Graph: `GetGrenadierComponent -> Bind Event to OnGrenadeTelegraph -> spawn decal at Landing, lifetime = TimeToImpact; bind OnGrenadeCancelled -> destroy it early`.
3. Compile, save.

Done when: camping >4s against a grenadier paints a ring on your cover roughly 3.5–4.5s before detonation (1s wind-up + flight + 2.5s fuse).

Gap 7: six BP children have no mesh/anim

1. For each of `BP_Enemy_Rusher/Heavy/Sniper/Officer/Grenadier/Shield`: Mesh component -> assign the soldier/mannequin skeletal mesh; Anim Class = the grunt's ABP.
2. Until distinct meshes exist, make 6 material-instance tints (e.g. Rusher red, Heavy dark, Sniper pale, Officer trim, Grenadier olive, Shield blue) so archetypes read at a glance — legibility is a design pillar.
3. Verify the skeleton has a `WeaponSocket` (right hand); without it the rifle attaches to the capsule root.

4. Heavy: consider mesh scale 1.1 until a real heavy mesh exists.

Done when: all seven types animate, hold the rifle correctly, and are tell-apart-able at 1500cm.

Gap 8+9+10: the level doesn't feed the systems yet

Without spawn zones the director never spawns (one warning log). Without SquadIds, placed enemies have **no squad behaviour at all** — no shared sightings, no flanking, no focus fire, no bounding (easy to misread as "AI broken"). Mission phases only change when level scripting says so.

1. Place 3+ AEnemySpawnZone actors at out-of-sight ingress points (behind buildings, stairwells, map edge); size the box, keep it over navmesh. ActivePhases empty = all phases.
2. On ONE zone set DirectorConfig = /Game/Enemy/AI/DA_DirectorConfig (or call SetDirectorConfig from the level BP at BeginPlay).
3. Level BP: objective trigger -> get EnemyDirectorSubsystem -> SetMissionPhase Objective; extraction trigger -> SetMissionPhase Extraction; optional alarm interactable -> TripAlarm.
4. Placed groups: select each enemy, set SquadId per group ("Alpha", "Bravo"...). Build one officer-led squad: 1 Officer + 3 Grunts sharing one id (bounding needs the officer plus >=2 grunt-archetype members in combat).
5. Each patrol enemy: assign its PatrolRoute actor (3+ points).
6. NavMesh bounds over patrols, cover, zones, flank ring (~900cm around likely player positions). Sniper arenas need unclaimed **stand-height** cover slots within 3000 of his perch-search origin and a long sightline.
7. Remember the cap trap: Infiltration MaxAlive is 8 and **placed enemies count** — a 10-enemy stealth level gets zero reinforcements until kills open headroom.

Done when: going Loud spawns a grunt pair within ~45s from off-screen; they arrive Searching toward you; two squads don't share sightings; the officer squad bounds.

Gap 11: no suppressed player weapon

bSuppressed, NoiseLoudness, NoiseRange exist on UWeaponDataAsset — no variant uses them. The stealth counter-loop (suppressed kills that don't alert the camp) is untestable.

1. Duplicate the player's rifle/pistol DA -> DA_<Weapon>_Suppressed: bSuppressed true, NoiseLoudness 0.15, NoiseRange 400 (vs 1.0/3000 unsuppressed).
2. Duplicate the weapon BP, assign the DA, add it to the player loadout path used in the test level.

Done when: an unsuppressed shot alerts enemies within 3000 (+30 suspicion each); the suppressed variant only nudges enemies within 400.

Gap 12: ragdoll check

1. Open the enemy skeletal mesh -> confirm a physics asset is assigned and its Ragdoll collision profile tumbles (kill one in PIE).
2. Takedown kills ragdoll 0.8s late by design (anim window) — that pause is normal.

2.2 — P1 polish (the game works without these; it doesn't feel finished)

- **Hit feedback** (design §8 "the player must see shots land"): bind `OnHitReact(EHitRegion)` to flinch montages (head gets its own), `OnSuppressedStateChanged(true)` to a duck/cower additive, `OnTakedownExecuted` to the enemy-side takedown reaction (you have 0.8s before ragdoll).
- **Heavy plates**: `OnPlateBroken(PlatesRemaining)` -> chunk burst FX + material swap per stage (3 plates -> 3 stages). This is also gameplay feedback: it tells the player frontal damage is working.
- **Rusher melee**: `OnMeleePerformed` -> swing montage + impact SFX.
- **Grenade**: body mesh + trail on `BP_EnemyGrenade`, explosion Niagara + SFX on detonate.
- **Heavy LMG variant**: duplicate rifle DA — `MagazineSize` 100, `BaseDamage` 18–20, `NoiseRange` 3200 — so his 2–3.5s bursts stop being mag-clipped at 3.0s and he reads as a weapons platform, not a grunt who holds the trigger.
- **Per-archetype bark sets**: duplicate `DA_Barks_Grunt` per type, re-voice lines (officer: commands; rusher: aggression; sniper: muttering). Assign in each `DA_Enemy_*` -> `Barks` -> `BarkSet`.
- **Bark SFX**: optional `Sound` per `FBarkDefinition` line set — subtitle text is the shipped contract, audio is gravy.

2.3 — P2 design-completeness (C++ — hand to a Claude Code session)

Item	Sketch
Grenade blast: noise + suppression + morale	<code>AEnemyGrenadeProjectile::Detonate:</code> <code>ReportNoiseEvent</code> (loudness 1.5, range 3500); radius-query AI pawns -> <code>RegisterNearMiss(2.0)</code> ; optional direct morale event so "nading his cluster dents squad morale" works without damage.
Officer passive morale floor	<code>USquadAuraComponent::Scan</code> also calls a <code>SetAuraFloor(25)</code> on members' morale comps (clear on leave/death) — design §7 "raises nearby allies' floor". DA field exists conceptually via <code>MoraleFloor</code> , currently 0 everywhere and never aura-driven.
Aura squad filter	Same scan: skip members whose <code>SquadId</code> differs from the officer's (today it buffs any enemy within 1500).
Outnumbered / reinforcements morale events	Squad-level headcount vs visible hostiles on the 1s morale tick (loss), <code>UEnemySquad</code> creation near an existing engaged squad -> gain broadcast. Both have DA-weight homes already (add two fields).
Lighting sight-fill factor	Needs a light-level probe (cheap: precomputed light volume samples or surface luminance at target) × DA factor — medium effort, design §4 lists it, P2 notes deferred it.
Side-aware flank bark	<code>BTask_EnemyFlank</code> knows the chosen ring point — compare its bearing vs target facing, pick "Flanking left!"/"right!" line variant (needs 2 extra bark lines too).
Post-search edginess	After Searching times out, hold a 30–60s "edgy" flag: suspicion fill ×1.5, patrol speed = combat speed. One float + timestamp in the awareness component.
Rusher vs suppression	Decision first: design says "barely suppresses" — current build = spread-only effect, charge never stops. If the companion-pins-the-rusher beat matters, add a brief stumble at full suppression; otherwise document as intended.
Delete <code>AEnemyBase</code>	Remove <code>Public/Enemy/EnemyBase.h</code> + <code>Private/Enemy/EnemyBase.cpp</code> (self-referenced only), delete any placed <code>BP_EnemyBase</code> , build.

3. Consolidated in-engine session (run top to bottom)

3.1 Data (10 min):

1. DA_Barks_Grunt: add GrenadeOut + Suppressing entries (lines above).
2. DA_Enemy_Grunt: DisplayName "Grunt".
3. Create DA_SniperRifle + BP_EnemySniperRifle; wire into DA_Enemy_Sniper.
4. Create suppressed player weapon DA + BP.
5. Optional: DA_LMG + BP for the heavy.

3.2 Blueprint bindings (the five delegates):

6. BP_Enemy_Sniper: OnLaserChanged -> laser beam on/off.
7. BP_Enemy_Grenadier: OnGrenadeTelegraph/OnGrenadeCancelled -> ground ring.
8. BP_Enemy_Shield: OnBoltOnComponentsReady -> shield component offset; DA_Enemy_Shield ShieldMesh -> real mesh.
9. BP_Enemy_Heavy: OnPlateBroken -> plate FX (P1).
10. All enemy BPs (or the shared ABP): OnHitReact, OnSuppressedStateChanged, OnTakedownExecuted, OnMeleePerformed -> montages (P1).

3.3 Characters:

11. Assign mesh + ABP + tint MI on the 6 new BP children; verify WeaponSocket; verify physics asset.

3.4 Level:

12. Spawn zones x3+, one carrying DA_DirectorConfig; phase triggers in level BP; PatrolRoutes; SquadIds (incl. officer squad); navmesh; stand-height cover for sniper arenas.

3.5 Smoke test (15 min, PIE): follow enemy_test_levels.md layouts; the per-phase QA lists live in enemy_code_plan.md. Minimum pass: grunt patrol->spot->cover->fire; rusher charge+melee; heavy front-shrug/rear-melt/slow-turn; sniper laser->shot->relocate; officer aura+focus+rally+bounding; grenadier 4s-camp->ring+bark->lob; shield block->break->fallback; officer death -> audible squad break; go-Loud -> director waves; suppression ducks a peeker.

4. Complete asset list

Characters & animation

Asset	Status	Used by	Notes
Soldier skeletal mesh (shared OK)	Missing on 6 BPs (grunt = mannequin)	all BP_Enemy_*	UE5 Quinn/Manny placeholder fine
6 tint material instances	Missing	per archetype	cheap archetype legibility until real meshes
ABP_Enemy (locomotion)	Exists for grunt — reuse	all	walk/run/crouch + aim offset
Physics asset (ragdoll)	Verify	all	death + takedown (delayed 0.8s)
WeaponSocket on skeleton	Verify	all	else rifle attaches to capsule

Asset	Status	Used by	Notes
Flinch montages (head + body)	Missing	OnHitReact (EHitRegion)	P1
Suppression duck/cower additive	Missing	OnSuppressedStateChanged	P1
Takedown reaction montage	Missing	OnTakedownExecuted	0.8s window before ragdoll
Melee swing montage	Missing	OnMeleePerformed (rusher)	P1
Death anims	Not needed	—	ragdoll covers it

Weapons

Asset	Status	Used by	Notes
DA_SniperRifle + BP_EnemySniperRifle	Missing — P0	DA_Enemy_Sniper	dmg 90, mag 5, noise 1.0/3500
Suppressed player weapon DA + BP	Missing — P0	stealth testing	bSuppressed, 0.15/400
DA_LMG + BP (heavy)	Optional P1	DA_Enemy_Heavy	mag 100, dmg 18–20
Riot shield static mesh	Placeholder (engine cube) — P0	DA_Enemy_ShieldShieldMesh	pivot at grip; offset via OnBoltOnComponentsReady
BP_Rifle / DA_AssaultRifle	Exists	all seven today	25 dmg, 10rps, 30 mag, noise 1.0/3000

VFX

Asset	Status	Used by	Notes
NS_SniperLaser beam	Missing — P0	OnLaserChanged	red beam muzzle->target, 2.0s window
Grenade warning ring (decal/Niagara)	Missing — P0	OnGrenadeTelegraph (Landing, ETA)	~350 radius = blast radius
Grenade body mesh + trail	Basic/missing	BP_EnemyGrenade	P1
Explosion Niagara + impact decal	Missing	grenade detonate	P1
Plate-break chunks/material stages	Missing	OnPlateBroken(N) (heavy)	P1, doubles as damage feedback
Muzzle flash / tracers	Exists	weapon Multicast_PlayFireFX	—

Audio (all optional — subtitles are the shipped contract)

Asset	Used by
Bark VO/SFX per FBarkDefinition.Sound	bark feed
Explosion SFX	grenade
Sniper aim hum (rising, 2s)	laser telegraph readability
Melee impact	rusher

Data edits (no new assets)

Edit	Pri
DA_Barks_Grunt += GrenadeOut, Suppressing	P0
DA_Enemy_Grunt DisplayName	P0
Per-archetype BarkSet DAs	P1

Level actors (placement, not assets)

Item	Notes
AEnemySpawnZone ×3+	off-sightline ingress; ONE carries DirectorConfig=DA_DirectorConfig
Mission-phase triggers (level BP)	SetMissionPhase Objective/Extraction; optional TripAlarm
APatrolRoute per patroller	3+ points
SquadId per placed group	incl. one officer + 3 grunts squad for bounding
NavMesh bounds	patrols, cover, zones, ~900cm flank ring around fight spaces
AICoverSlot coverage	sniper arenas need stand-height slots within 3000 + long sightlines
Test levels	layouts in enemy_test_levels.md (EnemyGym zones + ExtractionSlice)

5. Definition of done (P0)

Fix	Proven by
Bark lines + DisplayName	Grenadier wind-up and overwatch pin both print labelled subtitles
Sniper weapon + laser	Laser paints you 2.0s, then a shot that actually threatens; suppress him -> beam dies, no shot; he relocates after 2
Shield mesh/offset	Frontal blocked, 90° flank kills, grenade pops the plate
Meshes/ABP	Seven visually distinct enemies animating, rifles in hands
Zones/phases/SquadIds	Loud -> first wave ~45s from off-screen arriving Searching; officer squad bounds; "Bravo" keeps patrolling while "Alpha" fights
Suppressed weapon	Camp stays Calm through a suppressed kill at range; body discovery still escalates
Ragdoll	Deaths tumble; takedown ragdolls after the 0.8s anim window

Full per-phase QA checklists: enemy_code_plan.md (Phases 1–7) and enemy_gameplay_as_built.md §10 for the intended end-to-end feel.